in 2020 to 110.6 billion dollars by 2025, at a CAGR of 7.4 per cent.

According to Markets and Markets, the APAC region, which includes India, is forecast to hold the largest share of IIoT. But on the ground, the challenges are far greater than what these reports mention.

"Industry 4.0 sounds to me like teenage **x. Everyone thinks everyone else is doing it, but in reality, no one is doing it. There are a lot of things bundled under Industry 4.0. I don't think any of us can claim that we have achieved 100 per cent Industry 4.0. There is a long way to go for every one of us, and the road to Industry 4.0 is long," points out Sunil Banwari, COO, AT&S.

The coronavirus connection

A report published in McKinsey states that IIoT can help companies maintain a smooth flow of operation during the pandemic times. It is not a hidden fact that almost all the manufacturers, irrespective of the vertical they were in, were forced to shut down operations to make sure that humans working on shop floors do not become carriers of the virus.

In fact, as per the estimations made by the United Nations Conference on Trade and Development (UNCTAD), the Covid-19 outbreak could cause global FDI to shrink by five to fifteen per cent due to the downfall in the manufacturing sector coupled with factory shutdowns. These estimations, coupled with the decreasing demand for goods, sound like a perfect recipe for disaster in the manufacturing sector, which in turn will affect global GDPs and economies.

Several reports that mentioned the workforce is migrating back to their native villages added more fuel to the fire. It simply represents the fact that even if demand starts to rise again, manufacturers might not find workforce to compliment the same. Also, the cost of making arrangements for people to return from their native places to workplaces would be an added burden on manufacturers in the already tumbling economic reality.

India's GDP contracting by about 24 per cent in the June quarter is a sign that one shouldn't miss. All emerging countries, except China, witnessed GDP contractions in the range of five to nineteen per cent (Mint report). China's GDP, on the other hand, grew by almost three per cent even though coronavirus originated from this country.

The big picture

China's spending on smart manufacturing has surged year-on-year. In 2018, this investment shot up by 46 per cent and amounted to RMB 69.9 billion. Xi Jinping, president of China, had called for a robot revolution during a 2014 speech. The country since then has been aggressively investing in automating manufacturing processes. According to the International Federation of Robotics, Chinese companies installed 154,000 industrial robots in 2018, beating Japan at 55,200 and the US at 40,400.

"The topic of IIoT could not have been discussed at a better time than the time we are going through. Digital as an option has been picking pace in the last five to six months. Industry 4.0 is a very interesting area. Different manufacturing facilities and technologies are at different levels of maturity. We at Panasonic India are not only rolling out automation solutions for the manufacturing industry, but we are also deploying these at a lot of Panasonic facilities in India," shares Manish Misra, CIO, Panasonic India.

Vinod Sharma, MD, Deki Electronics, mentions that his company (an SME) has been looking at Industry 4.0 as a blind man looks at an elephant. He says, "IIoT technologies can help in bringing down costs to some extent. These could be via preventive maintenance, predictive maintenance, and by increasing productivity.''

"IIoT gives us an analytical framework to dwell at a place. Our finding out of IIoT is that when big companies

Advantages of Industry 4.0

- Brings down overall costs involved in manufacturing as well as in supply verticals.
- Enables predictive and preventive maintenance. The same can also help in increasing productivity.
- Helps in comparing machines and hence choosing the right one for the right job.
- Leaders can develop efficient workflow management as well as schedule shop floor shifts in a better way.
- Helps in pointing towards the average manufactured components or final goods. This can help suppliers gain more confidence with their customers.
- Enables real-time data sharing with manufacturers and their customers. This can help in keeping the manufacturing process in control and making sure that the goods being manufactured are up-to-the quality as set by the customer.

The challenges

- Data collection, data extraction, turning data into information, and turning the information to analysis represent the biggest challenges associated with Industry 4.0.
- Often the outlay of projects related to industry 4.0 is so large that a lot of mid-size companies cannot ascertain whether to go for the same or not. Providers of IIoT/Industry 4.0 solutions might need to tailor-make such solutions for mid-sized companies.
- Even if a small project involving automation goes wrong, it might impact the working of the whole team. This, in many cases, can make the management think that industry 4.0 does not work.
- SMEs involved in the manufacturing sector may find it difficult to compete in the market as other companies start adopting IIoT. The costs involved in the manufacturing process might get lowered but the initial investment costs might be very high for a lot of SMEs.
- IIoT lacks proper standardisation protocols. Similarly, at times it is difficult to obtain a complete set of data when it comes to machine to machine connections.
- Implementing IIoT in legacy machines can be a big challenge as these were not made keeping in mind the connected and automated technologies.
- The cost of implementing IIoT in a mix of legacy machines and the modern ones might go up exponentially.